

Question:

Stephen placed an online order for groceries. His final bill came to \$40.00. Because this was through a delivery vendor, they tacked on a 25% fee to his final total and charged him \$3.00 in delivery fees. Stephen also added a \$4.00 tip. After the extra fees, what was the final price of Stephen's groceries?

Reasoning + Answer:

Let's carefully break down the problem step by step to find the final price of Stephen's groceries after all fees and tips .

We are told :

- Stephen's **final bill** came to **\$40.00** .
- The delivery vendor added a **25 % fee** to the final total .
- They also charged a **\$3.00 delivery fee** .
- Stephen added a **\$4.00 tip** .

We are to find the **final price** of the groceries after all extra fees and the tip .

But the key is to understand what the \$40.00 represents .

It says : " His final bill came to \$40.00 " âĢĲ and then it says " because this was through a delivery vendor , they tacked on a 25 % fee to his final total and charged him \$3.00 in delivery fees ."

This suggests that the \$40.00 is the **original grocery cost before fees** , and then fees are added on top .

LetâĢĲs re-express the situation :

- Let the original cost of groceries be **\$40.00** .
- Then , a **25 % fee** is added to the final total .
- Also , a **\$3.00 delivery fee** is added .
- And Stephen adds a **\$4.00 tip** .

Wait âĢĲ the phrasing is : " they tacked on a 25 % fee to his final total " âĢĲ that is ambiguous .

LetâĢĲs re-read : âĢĲ His final bill came to \$40.00 . Because this was through a delivery vendor , they tacked on a 25 % fee to his final total and charged him \$3.00 in delivery fees .âĢĲ

So the **final bill** is \$40.00 âĢĲ that is , the total amount Stephen was charged .

But then it says they tacked on a 25 % fee to his final total âĢĲ that would mean the \$40.00 includes the 25 % fee ?

But then it says they charged \$3.00 in delivery fees âĢĲ which would be in addition ?

So this is confusing .

LetâĢĲs try to interpret it logically .

Possibility : The \$40.00 is the **grocery cost** .

Then :

- Delivery vendor adds a 25 % fee on top of the grocery total .
- Also charges \$3.00 delivery fee .
- Stephen adds \$4.00 tip .

Then total final price = grocery cost + 25 % fee + delivery fee + tip .

So :

Grocery cost = \$40.00

25 % fee on \$40.00 = 0.25 Ã 40 = \$10.00

Delivery fee = \$3.00

Tip = \$4.00

Total final price = 40 + 10 + 3 + 4 = \$57.00

But the problem says : âĢĲ His final bill came to \$40.00âĢĲ âĢĲ that would mean that \$40 is the total after fees , not before .

So perhaps the \$40.00 is the total amount Stephen paid âĢĲ including the 25 % fee and delivery and tip ?

But then it says : âĢĲ they tacked on a 25 % fee to his final totalâĢĲ âĢĲ which would mean the \$40.00 includes the 25 % fee .

And also charged \$3.00 delivery fee âĢĲ so that would be part of the \$40 ?

And \$4.00 tip ?

But then it says : âĢĲ After the extra fees , what was the final price of Stephen's groceries ?âĢĲ

So the final price of the groceries âĢĲ that is , the original grocery price âĢĲ or the total amount Stephen paid ?

It says : âĢĲ what was the final price of Stephen's groceries ?âĢĲ

This is ambiguous .

But in common usage , âĢĲ final price of groceries